SMARTRPD · 2D ANNOTATION

Jaw Design Auto-Placement

The clinical reasoning behind **Load Template Jaw** — how each arch is classified, and how that class chooses the major connector, the rest seats, the clasps and the bars that are placed on it.

Scope Rules only — no implementation detail **Modules** JawDesign · JawDesignProposal **Date** 17 August 2026

1 How the proposal is produced

Nothing is uploaded and nothing is asked. The case's saved design is already loaded into the arch when the page opens, so pressing **Load Template Jaw** reads the arch exactly as it stands — including presence edits made since opening — and reasons forwards from it.

STEP 1	STEP 2	STEP 3	STEP 4
Classify	**Choose**	**Preview**	**Confirm**
Each arch separately, from tooth presence alone.	Connector, rests, clasps and bars from that class.	The design is drawn, thumbnailed, then rolled back.	Tick either arch, or both, and place.

The preview is the real design, not an illustration: the proposal is applied to a snapshot of the arch, the arch is rendered, both thumbnails are captured, and the snapshot is then restored. What the thumbnail shows is exactly what **Place Design** produces.

Each arch is decided independently and can be placed independently. An arch that Kennedy gives no class — a complete arch, or a fully edentulous one — is greyed out and cannot be ticked: there is nothing to replace in the first case, and no abutment to carry a framework in the second. Placing one arch never touches the other.

The proposal replaces the arch's design, it does not add to it. Whatever was on a ticked arch is stripped before the new design lands, so an old connector or clasp can never be left drawing over the proposed one. Tooth presence and status survive untouched — they are what the proposal was derived from — and an arch with nothing to propose keeps the design it already has.

> **TWO STANDING CONSTRAINTS**
>
> **No millimetres.** The 2D arch measures nothing, so every rule whose trigger is a dimension — floor-of-mouth depth, ridge resorption, undercut depth, rest-seat preparation — is dropped, and the primary choice for that situation is taken. **No survey line.** Undercut location and survey-line shape are unknown, so any rule that depends on them is left as a manual decision.

2 Step one — Kennedy classification

The classifier reads only which teeth are present. It walks the arch from one terminal molar, through the midline, to the other — so the two ends of that walk are the two ends of the arch.

A. WHICH SLOTS COUNT

A **missing third molar is dropped from the arch before anything is measured** (Applegate rule 2) — it is not being replaced, so it neither creates a space nor opens the back of the arch. A **present** third molar counts normally and can serve as an abutment (rule 3).

B. EDENTULOUS SPANS

Every maximal run of consecutive missing teeth becomes one span. Each span records four things:

- **Distal extension** — the span reaches an end of the arch, so no tooth remains behind it on that side. Everything else is **tooth-bounded**.
- **Crosses the midline** — the span occupies both quadrants.
- **Abutments** — the standing tooth at each closed end, together with *which of its own surfaces faces the space*. That surface is what the rest and clasp rules consume later.
- **Sides** — patient's right or left, taken from the FDI quadrant.

CLASS DECISION — THE MOST POSTERIOR AREA DECIDES (APPLEGATE RULE 5)

RESULT	CONDITION	READING
Class I	Distal-extension spans on **both** sides	Bilateral free-end
Class II	Distal extension on **one** side	Unilateral free-end
Class IV	Exactly **one** span, and it crosses the midline	Single anterior space
Class III	Everything else	All spans tooth-bounded
Dentate	No spans at all	No class; nothing to design
Edentulous	Every counted slot missing	No class; no abutments

Only the two ends of the arch can be open, so there can never be more than two distal extensions — which is why the first two rows are exhaustive tests rather than approximations.

C. MODIFICATIONS

Every span the class did not itself consume is counted as a modification (rules 6 and 7): Class I consumes two spans, Class II one, Class III one. **Class IV never takes a modification** (rule 8) — a midline-crossing anterior space alongside any other space is not Class IV at all, it falls through to Class III with the extra spans counted. The label reads *Class II mod 1*, and so on.

> **KNOWN LIMITATION — APPLEGATE RULE 4**
>
> Rule 4 says a missing tooth that will *not* be replaced is not counted. The arch carries no "will replace?" flag per tooth, so every missing tooth other than a third molar is counted. An arch where a space is deliberately being left unrestored will classify one class too aggressively.

3 Step two — the major connector

One connector per arch, chosen from the class first and then adjusted for two conditions the arch does record. The material decides which table applies: cast metal takes the selection matrix below, full acrylic takes its own — acrylic is a different prosthesis, mucosa-borne, so it spreads the bite across the soft tissues instead of running a rigid connector between abutments.

CAST METAL — PRIMARY CHOICE BY CLASS

CLASS	MAXILLA	MANDIBLE	WHY
Class I	A-P Strap	Lingual Bar	Bilateral extension needs cross-arch rigidity above; below, the bar is standard while the anteriors are sound.
Class II	A-P Strap	Lingual Bar	A one-sided extension loads the arch asymmetrically, so rigidity still matters.
Class III	Palatal Strap	Lingual Bar	Fully tooth-borne — coverage stays minimal, and there is no rotation to resist.
Class IV	A-P Strap	Lingual Plate	Anterior replacement resists displacement above; below, the plate splints the anterior abutments and provides indirect retention.

TWO OVERRIDES THAT OUTRANK THE CLASS

TRIGGER	BECOMES	REASONING
Any span abutment marked *compromised* (maxilla)	Palatal Plate	Full palatal coverage spreads load off an abutment that cannot carry it.
Any present, compromised lower anterior	Lingual Plate	Splinting weakened anteriors is the reason the lingual plate exists.

Neither override applies where the connector is already a plate. Compromised status is read from the tooth itself, not inferred.

FULL ACRYLIC — ITS OWN TABLE

SITUATION	MAXILLA	MANDIBLE
Free-end saddle (Class I or II)	Palatal Plate	Lingual Plate
Tooth-bounded, fewer than 5 teeth to replace	Horse shoe	Lingual Plate
Tooth-bounded, 5 or more teeth to replace	Palatal Plate	Lingual Plate

A free-end saddle needs the plate's tissue coverage to carry the bite. A short bounded case is light enough to leave the palate open. The count is taken from the classified spans, so a missing third molar that rule 2 already dropped is not held against the arch. There is no lower horseshoe, so below, the lingual plate answers every class.

DOWNSTREAM EFFECT

The connector determines the plating: a plate, strap or horseshoe plates every tooth it covers, while a lingual bar plates none. Two display states also follow from it — the A-P strap opens the palatal window on the maxillary overlay, and the lingual bar hides the lower plate visuals. Each is applied only to the arch that was actually placed.

4 Step three — rest seats

Rests decide where the denture is supported and, on a free-end case, where it pivots. Three rules run in order, and their results are merged.

REST PLACEMENT

RULE	WHERE	PURPOSE
A · Bounded span	On each abutment, on **the surface facing the space**.	Support is taken directly across the space at both ends — the classic tooth-borne case.
B · Free-end abutment	On the terminal abutment's **mesial**, i.e. away from the saddle.	Moves the fulcrum forward so the saddle rotates *away* from the abutment under load instead of levering it out of its socket.
C · Indirect retainer	As far from the fulcrum line as the arch allows — see below.	Resists the saddle lifting away from the ridge on the other side of the pivot.

WHERE THE INDIRECT RETAINER GOES

CLASS	PLACEMENT
Class I	Both sides — first premolar, or the canine's cingulum if the premolar is gone.
Class II	The side **without** the extension only; the extension side has no fulcrum to lever against.
Class III	None — fully tooth-supported, so there is no rotation to control.
Class IV	The fulcrum is anterior, so the retainer goes *behind* it: the posterior-most standing tooth of each quadrant.

THREE QUALIFIERS

- **Anteriors take a cingulum rest**, not an occlusal one — incisors and canines are seated on the lingual shelf. Incisal rests, the documented fallback for a canine that cannot take a cingulum rest, are never proposed: enamel thickness is not recorded anywhere in the 2D arch.
- **A tooth already carrying a primary rest is skipped** when the indirect rests are assigned — it is already doing that job.
- **Duplicates are removed per surface, not per tooth.** A tooth standing between two spans legitimately takes a rest on each side, so only an identical tooth-and-surface pair is dropped.

5 Step four — clasps

Each abutment gets **one retentive assembly, and only one.** The abutments of every span are visited in turn and the first rule that fits claims the tooth; a tooth already claimed is never reconsidered.

SELECTION LADDER — FIRST MATCH WINS

PRIORITY	SITUATION	ASSEMBLY	WHAT IS PLACED
1	Abutment of a **free-end** span	RPI, else RPA	See §6 — an I-bar with its proximal plate where the bar is indicated and can be resolved, otherwise a buccal Akers arm off the same plate.
2	**Isolated molar** (no standing neighbour either side)	Ring clasp	A ring encircling to a mesial undercut, plus a proximal plate. Buccal in the maxilla, lingual in the mandible — the side each jaw's molars tip toward.
3	Abutment of a **bounded** span, in the aesthetic zone	I-bar	Where a bar surface resolves — see §6.
4	Any other bounded abutment	Circlet	Retentive arm and reciprocal arm, both on the corner **away** from the space, buccal and lingual respectively.

THE CIRCLET CONVENTION

A circlet is rested against the space and its arms run to the far corner of the tooth, buccal for the retentive arm and lingual for the reciprocal one, so the two neutralise each other. This mirrors the Simple Circum assembly the catalog itself places when a rest is picked by hand: rest mesial → arms distal, rest distal → arms mesial.

CLASS II BRACING

A Class II arch has retention on the extension side but nothing holding the opposite quadrant, so that quadrant is braced with an **embrasure (double Akers) pair**: the two adjacent standing posterior teeth furthest back, clasped as a pair across their shared embrasure, each rested against it. The pair brings its own two rest seats, which is why they appear in the rest list as *bracing* rests. If the quadrant has no adjacent posterior pair left unclaimed, no pair is placed.

> **WORTH A CLINICAL REVIEW — EMBRASURE ARM POSITION**
>
> As currently implemented, both teeth of the embrasure pair carry their rest *and* their arms on the shared-embrasure corner, so the two retentive terminals converge at the embrasure. The conventional double Akers rests in the embrasure but extends the retentive terminals *away* from it — which is also the convention every other clasp here follows (arms opposite the rest). If the outward form is the one you want, this is a one-line change in the pair builder.

TERMINAL MOLARS ARE NEVER CHOSEN

The four terminal molars are excluded from any site the proposal picks *for itself* — the embrasure pair and the Class IV indirect retainer — matching the exclusion the arch already applies to automatic mesh and plating. A third molar that genuinely bounds a space is still treated as an abutment: that is not a choice, it is the anatomy.

6 Step five — bars

Only the **I-bar** is ever proposed, and always as part of an assembly rather than on its own. Two questions decide it: is the bar indicated on this tooth, and can a bar surface be resolved against the arch?

QUESTION	ANSWER USED
Indicated on a free-end abutment?	Canine, first or second premolar only. This is the RPI system's own indication: the I-bar approaches the tooth from below, so it disengages under load rather than levering the abutment. A molar terminal abutment falls back to RPA.
Indicated on a bounded abutment?	Anteriors and premolars — the aesthetic zone, where a suprabulge arm would cross visible facial enamel. Molars keep the circlet.
Can a surface be resolved?	Only against a saddle **directly adjacent** to the tooth. A bar bases from the denture saddle, so with none next door there is nothing for the arm to rise from — and the circlet or RPA is used instead.

The bar surface is always **derived, never chosen.** The arch scans outward from the tooth for a meshed neighbour, preferring the distal side and the nearer of two candidates, and returns a surface label that is keyed to per-tooth geometry tuning rather than to anatomy — the label is inverted relative to the saddle it bases from. The proposal takes that label as given, and only checks that it came from an *adjacent* saddle before accepting it.

> **PLANNING AGAINST THE FINISHED ARCH**
>
> Clasps are planned before anything has been placed, so at that moment no saddle is meshed and no bar surface would ever resolve — every free-end abutment would silently fall back to RPA. The planner therefore reasons against a *projection* of the arch with the saddles already meshed, exactly as the placement step will build it.

T-BAR AND Y-BAR ARE NEVER PROPOSED

Their indications are the position of the undercut and the shape of the survey line. The 2D arch records neither, so choosing between them would be a guess. They remain manual picks from the catalog.

7 What gets placed, and in what order

The order is not cosmetic — each step depends on the one before it, and it is the same order the catalog follows when a designer clicks through by hand.

ORDER	STEP	WHY HERE
1	Strip the placed arches	Every component already on them, so nothing from the old design is left overlapping the new one. Presence and status are kept. Arches with no proposal are skipped entirely.
2	Mesh on every saddle	Only the arches being placed. A connector can only be placed on teeth that anchor it, and a bar needs the saddle to base from.
3	Proximal plate on every standing tooth	The connector switch expects the plating to already exist.
4	Switch to the chosen major connector	—
5	Re-sync plating to that connector	A plate, strap or horseshoe keeps the plating it covers; a lingual bar strips it again.
6	Rest seats	Nothing downstream reads them.
7	Clasps and bars	Last, so that a clasp's own proximal plate survives the plating sync above.

Two placement invariants hold throughout. A tooth reciprocates with **one** element only — a reciprocal clasp, a proximal plate or a cross-mesh plate, never two at once, so placing one clears the others. And nothing is ever placed twice: an identical component-and-surface pair already on the tooth is left alone.

8 Deliberately not modelled

LEFT OUT	BECAUSE
Every millimetre trigger	Floor-of-mouth depth, ridge resorption, rest-seat dimensions, undercut depth. The 2D arch measures nothing, so the primary choice for the situation always wins.
Undercut position and survey line	Unknown. This is what rules out the T-bar and Y-bar, and any rule keyed to where retention sits on a tooth.
Tori and tissue anatomy	Not recorded anywhere in the 2D model.
Applegate rule 4	Needs a per-tooth "will replace?" flag the arch does not carry (see §2).
Incisal rests	Their trigger is enamel thickness on a canine, which is not recorded.

The alternatives that *are* knowable do switch the choice: a compromised abutment, a compromised lower anterior, the number of teeth to replace, whether a molar is isolated, and whether a saddle sits next to a given tooth.

9 Worked examples

Bilateral free-end, cast metal

Lower · 36 37 38 46 47 48 missing

Class	Class I — 38 and 48 dropped, both ends open
Connector	Lingual Bar
Rests	Mesial on 35 and 45 (free-end); indirect mesial on 34 and 44
Clasps	RPI on 35 and 45 — both premolars with an adjacent saddle

Unilateral free-end, cast metal

Upper · 25 26 27 28 missing

Class	Class II — 28 dropped, left end open
Connector	A-P Strap
Rests	Mesial on 24; indirect mesial on 14; bracing rests from the embrasure pair
Clasps	RPI on 24; embrasure pair bracing the right quadrant

Anterior span

Upper · 12 11 21 22 missing

Class	Class IV — one span, crossing the midline
Connector	A-P Strap
Rests	Cingulum rests on 13 and 23; indirect posteriorly on 17 and 27 (18/28 not chosen)
Clasps	Canine abutments — bar clasp where the adjacent saddle resolves one

Short bounded span, full acrylic

Upper · three teeth missing, not free-end

Class	Class III
Connector	Horse shoe — bounded and fewer than five teeth, so the palate stays open
Rests	On each abutment, facing the space
Clasps	Circlets running away from the space, unless the abutment is in the aesthetic zone

Isolated molar

Lower · 35 and 37 missing

Class	Class III mod 1 — both spans bounded
Clasps	Ring clasp on 36 with a proximal plate; the ring engages the mesio-lingual, the side lower molars tip toward

Compromised abutment

Upper Class III with a failing abutment

Connector	Palatal Plate, not the Palatal Strap the class would give — full coverage spreads load off the abutment
Rests	Unchanged; the override affects the connector only